GlobalGeoTree Berlin: Observations, Sources, and District Distribution

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1 Purpose

This report focuses only on **Berlin**. It answers three questions:

1. How many GlobalGeoTree observations and recorded species are available for Berlin?
2. Which data sources contribute Berlin observations?
3. How are tree observations distributed across Berlin districts?

The district-level analysis requires Berlin district polygons. The report first tries to read a local GeoJSON file. If no local file exists, it tries to download a simple Berlin district GeoJSON. If neither option works, the report falls back to a Berlin bounding-box analysis and shows a point-density map without district borders.

2 Setup

3 Load observations

Input data summary

item	value
Input file	GlobalGeoTree.csv
Rows in input file	6,263,345
Rows after coordinate/species cleaning	6,263,345
Countries after cleaning	221

4 Preview of the dataset

The table below shows the first rows of the cleaned observations file. This is useful for checking column names and whether fields such as `source`, `year`, `longitude`, and `latitude` are available.

First 10 cleaned rows

sample_id	country_code	level0	level1_family	level2_genus	level3_species	location	source	speci
1	MW	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	Malawi	iNaturalist Research-grade Observations	5,5
2	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes formosissimus	Mozambique	iNaturalist Research	5,5

		Broadleaf			formosissimus		Research-grade Observations	
3	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	Mozambique	iNaturalist Research-grade Observations	7,2
4	MZ	Evergreen Broadleaf	Acanthaceae	Anisotes	Anisotes sessiliflorus	Mozambique	iNaturalist Research-grade Observations	7,2
5	TZ	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia marina	Tanzania, United Republic of	iNaturalist Research-grade Observations	2,9
6	KE	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Kenya	naturgucker	2,9
7	NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Biodiversity of Some Coastal Communities in Akwa-Ibom State, Nigeria	2,9
8	NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Biodiversity of Some Coastal Communities in Akwa-Ibom State, Nigeria	2,9
9	NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Flora of Coastal Environment in Cross-River State, Nigeria	2,9
10	NG	Evergreen Broadleaf	Acanthaceae	Avicennia	Avicennia spec	Nigeria	Flora of Coastal Environment in Cross-River State, Nigeria	2,9

5 Load Berlin district boundaries

Berlin district boundary status

item	value

item	value
District boundary status	Downloaded district boundaries from https://raw.githubusercontent.com/codeforgermany/click_that_hood/main/public/data/berlin.geojson
Number of district polygons	97

6 Filter observations to Berlin

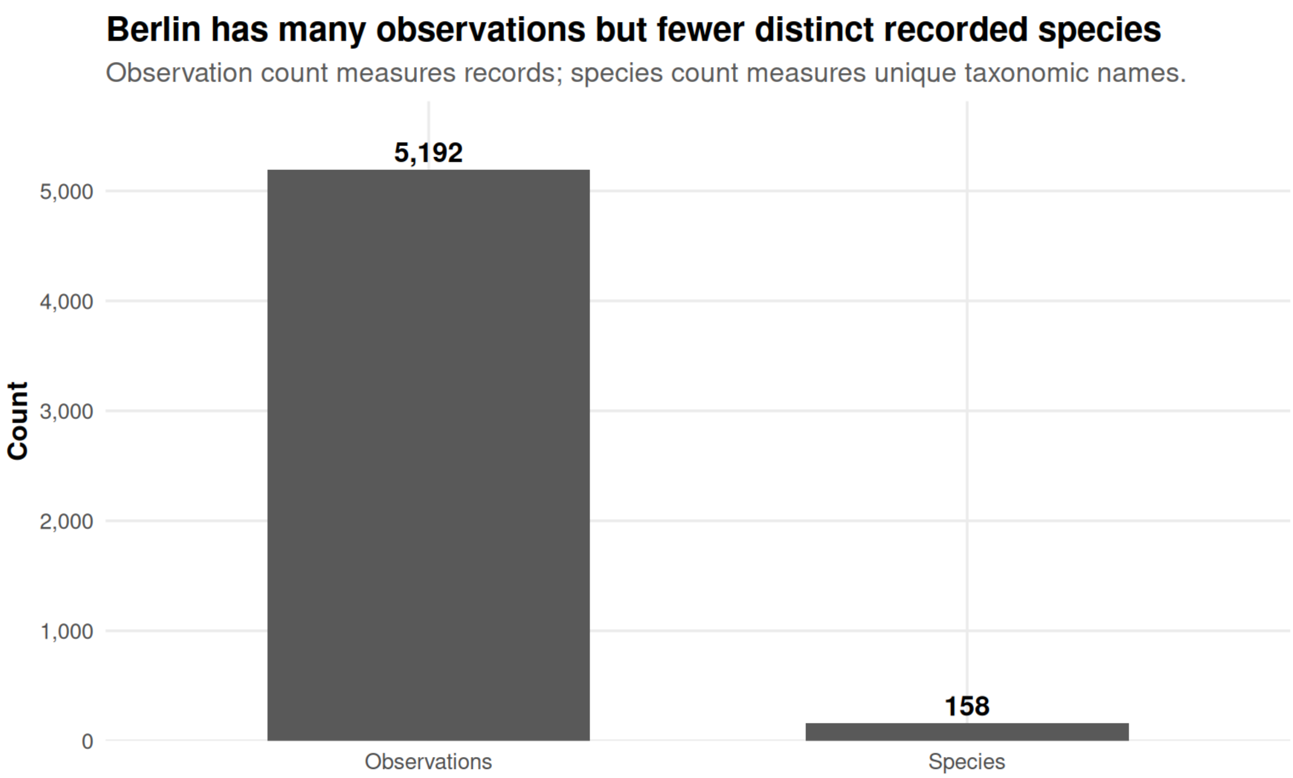
If district polygons are available, Berlin observations are identified by spatially joining Germany observations to Berlin district polygons. This is the preferred method because it avoids relying on text fields such as `location`.

If district polygons are not available, the report uses an approximate Berlin bounding box. This fallback is useful for a quick check, but it is less precise than district polygons.

Berlin observation summary

metric	value
Berlin filtering method	Spatial join to Berlin district polygons
Berlin observations	5,192
Berlin recorded species	158
Available source values	13
Available years	2015–2024

7 How many observations and species are available for Berlin?

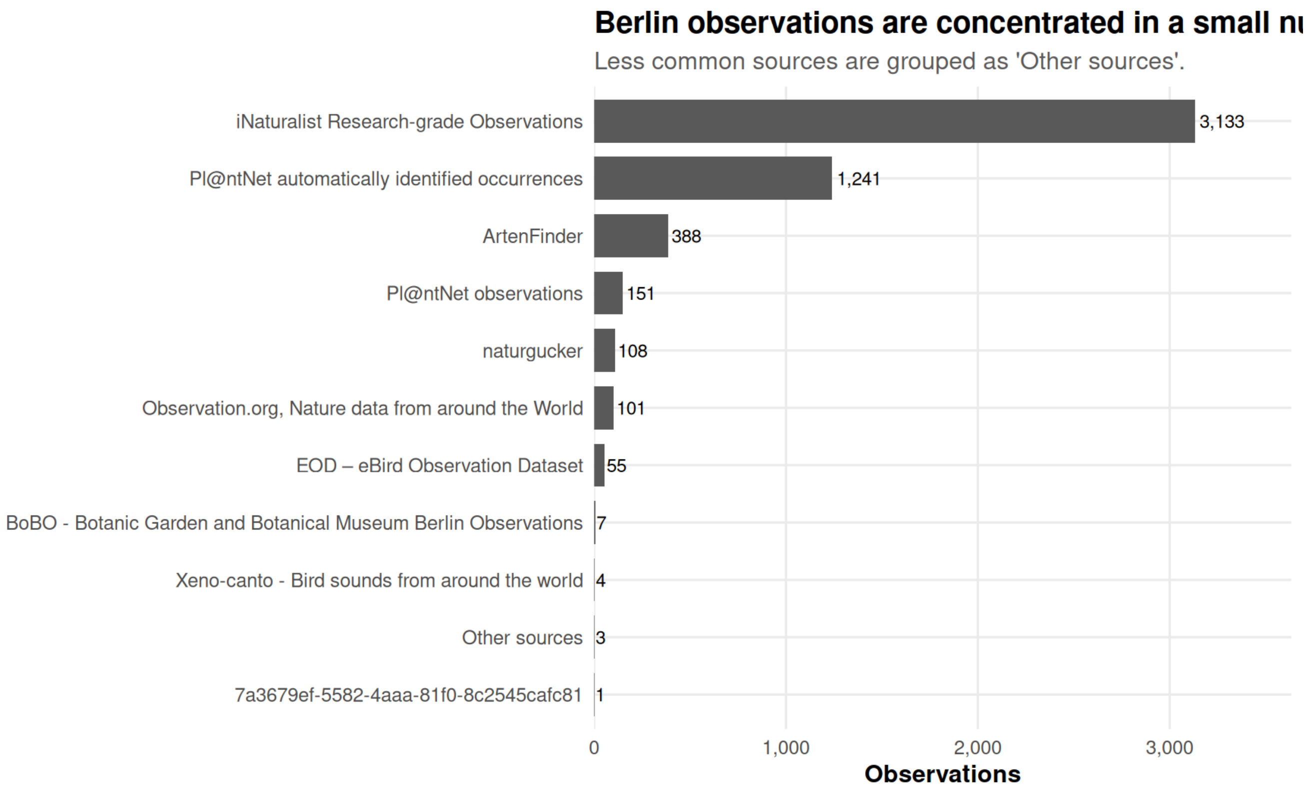


Berlin observation and species totals in the selected GlobalGeoTree file.

8 What are the sources of Berlin observations?

Top Berlin data sources

source	observations	share
iNaturalist Research-grade Observations	3,133	60.3%
Pl@ntNet automatically identified occurrences	1,241	23.9%
ArtenFinder	388	7.5%
Pl@ntNet observations	151	2.9%
naturgucker	108	2.1%
Observation.org, Nature data from around the World	101	1.9%
EOD – eBird Observation Dataset	55	1.1%
BoBO - Botanic Garden and Botanical Museum Berlin Observations	7	0.1%
Xeno-canto - Bird sounds from around the world	4	0.1%
7a3679ef-5582-4aaa-81f0-8c2545cafc81	1	0.0%
Database for alien invasive plants occurrences in Germany	1	0.0%
Database of field studies on environmental impacts of invasive plant species in Europe	1	0.0%
Spatially explicit database of tree related microhabitats (TreMs)	1	0.0%



The largest sources shape where Berlin tree observations appear in the dataset.

9 Spatial distribution of Berlin observations

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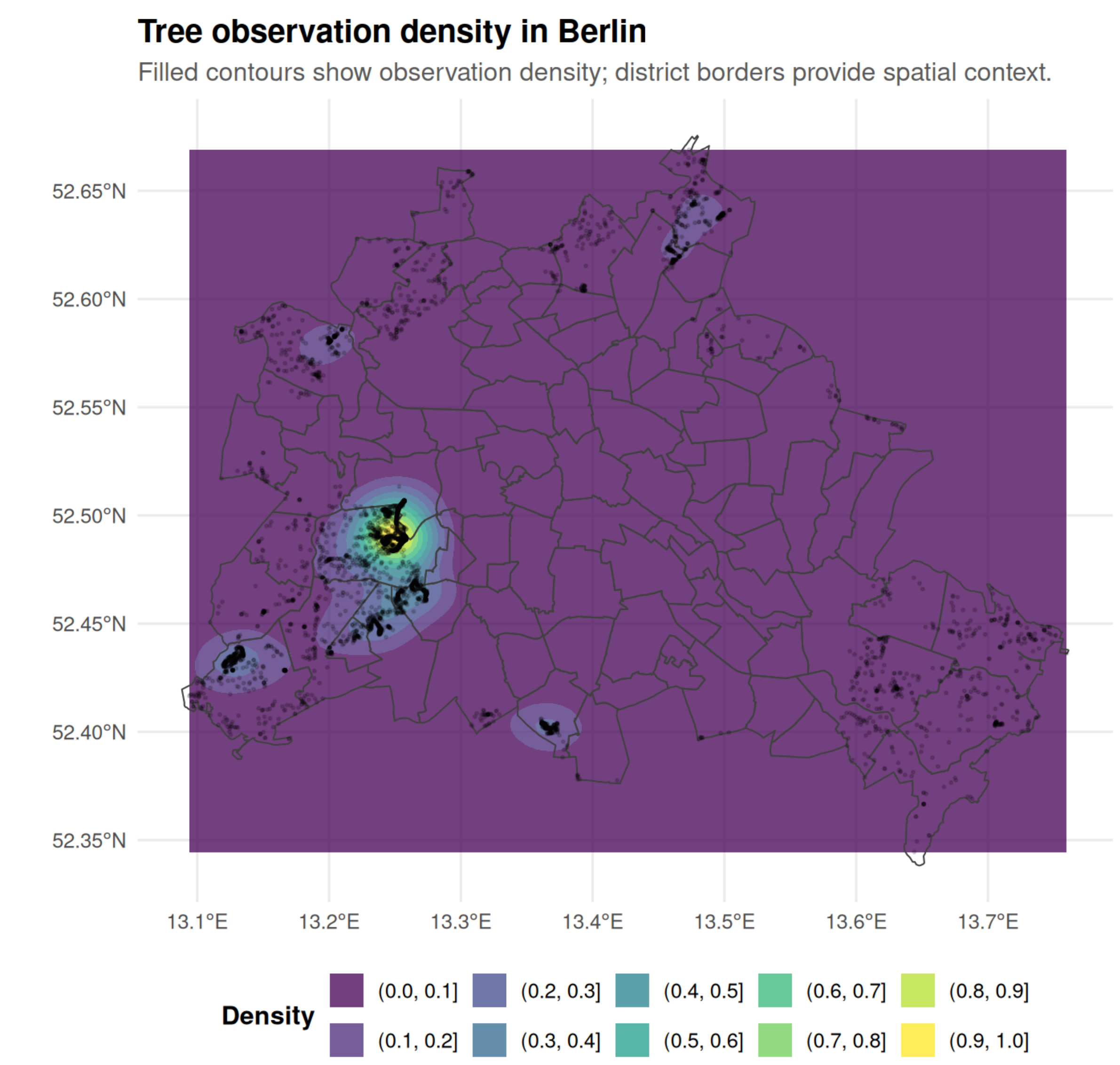

The density map shows where observations are concentrated inside Berlin. If district boundaries are available, they are drawn on top of the density layer.

10 Tree distribution by Berlin district

Observation, species, and source counts by Berlin district

district	observations	species	sources
Grunewald	1,714	87	7
Wannsee	490	76	8
Dahlem	298	39	7
Zehlendorf	274	43	6
Hakenfelde	273	46	8
Marienfelde	261	49	6
Buch	247	60	7
Westend	165	32	5

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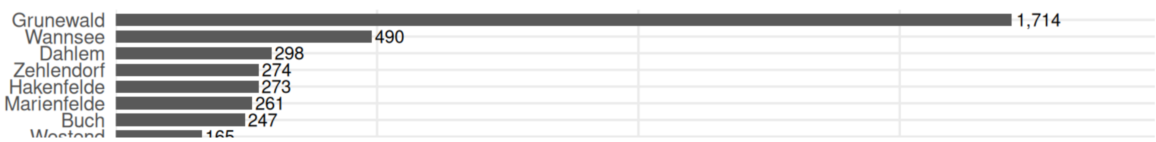
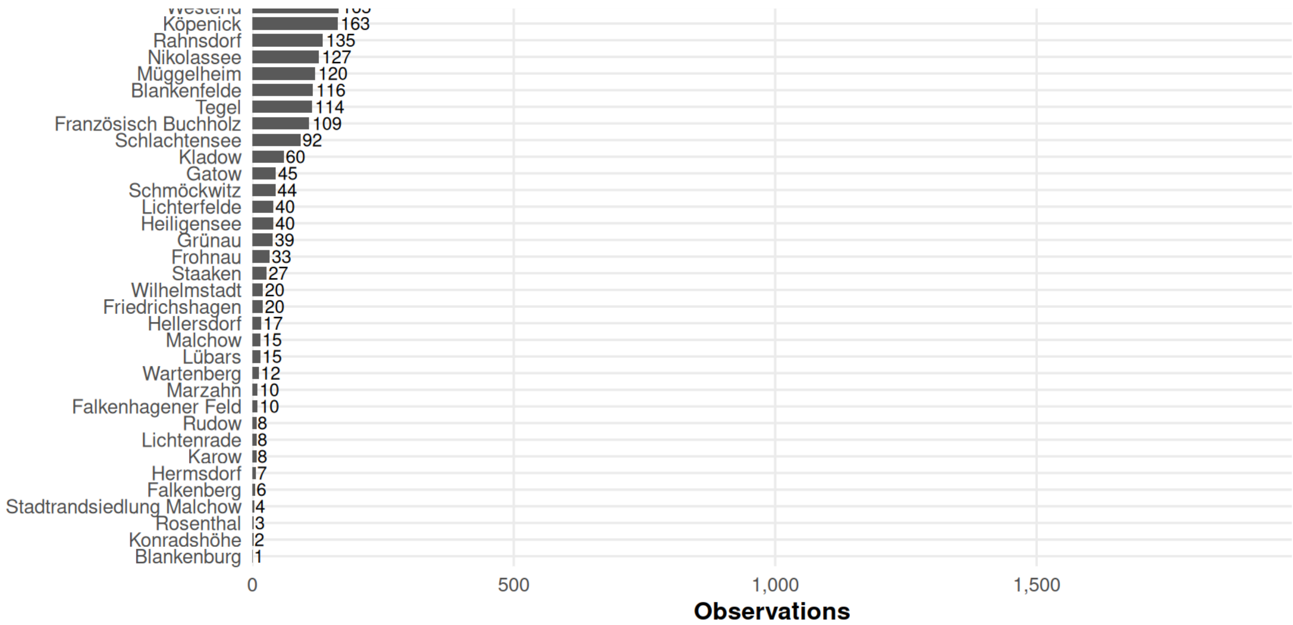
Köpenick	163	45	7
Rahnsdorf	135	36	5
Nikolassee	127	43	7
Müggelheim	120	39	8
Blankenfelde	116	41	7
Tegel	114	29	8
Französisch Buchholz	109	35	5
Schlachtensee	92	22	4
Kladow	60	23	5
Gatow	45	24	5
Schmöckwitz	44	24	7
Heiligensee	40	15	5
Lichterfelde	40	31	8
Grünau	39	18	6
Frohnau	33	22	5
Staaken	27	15	6
Friedrichshagen	20	11	6
Wilhelmstadt	20	12	4
Hellersdorf	17	8	4
Lübars	15	11	3
Malchow	15	9	3
Wartenberg	12	10	4
Falkenhagener Feld	10	9	2
Marzahn	10	8	3
Karow	8	8	2
Lichtenrade	8	6	2
Rudow	8	6	1
Hermsdorf	7	5	2
Falkenberg	6	5	2
Stadtrandsiedlung Malchow	4	3	2
Rosenthal	3	3	2
Konradshöhe	2	2	1
Blankenburg	1	1	1

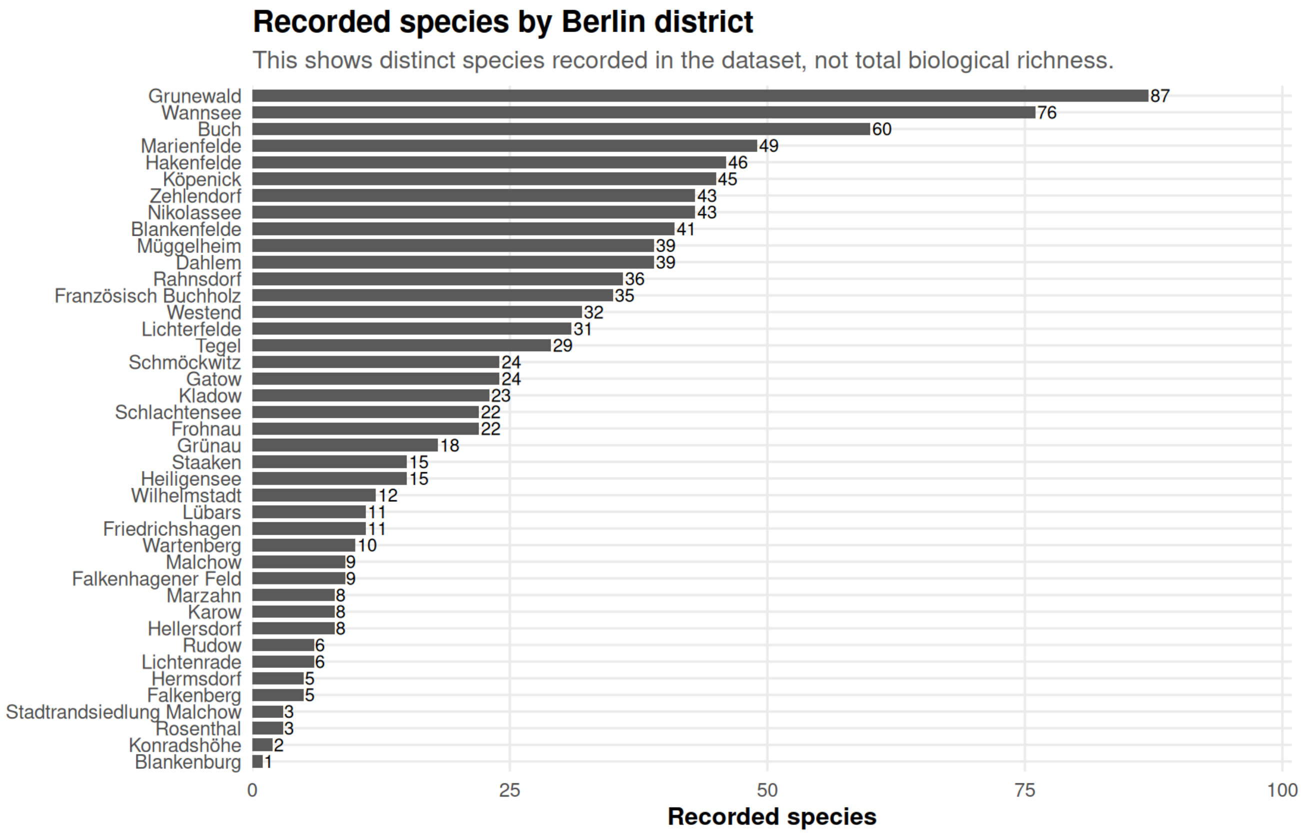
Berlin observations by district

Higher values can reflect tree presence, recording effort, or both.

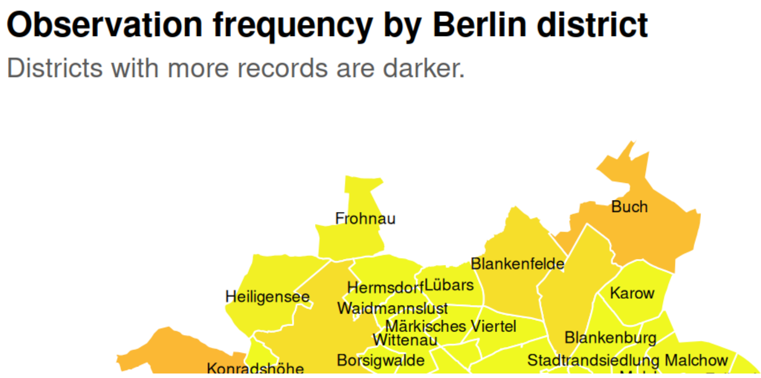


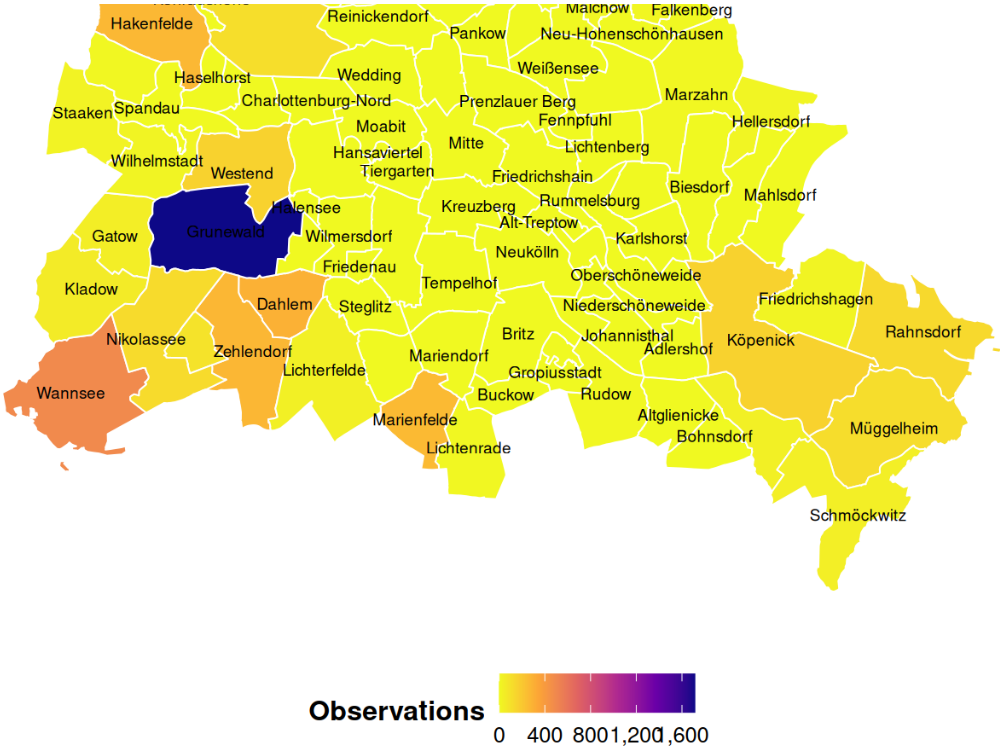


District-level observation counts show whether records are evenly distributed across Berlin or concentrated in specific districts.



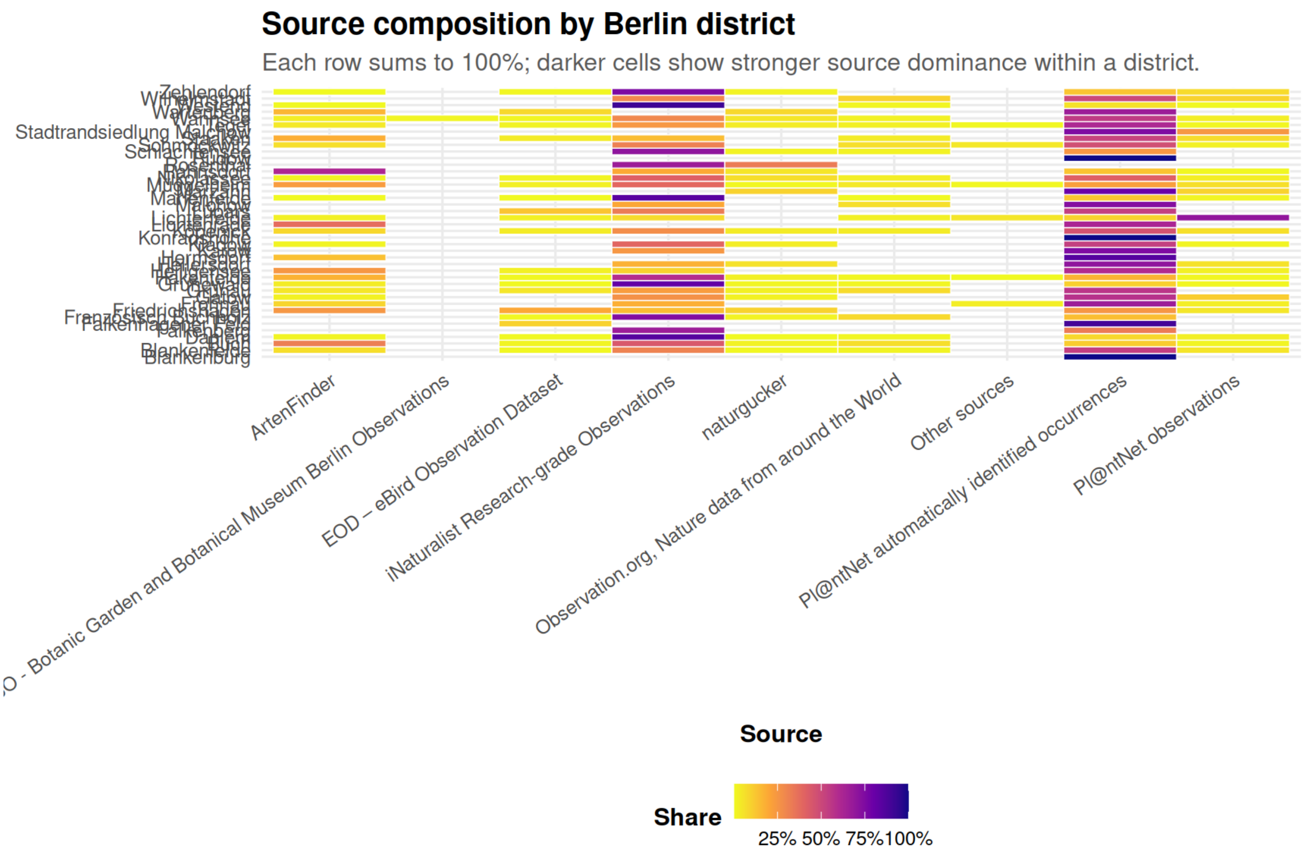
Species counts by district summarize taxonomic richness observed in each district.





The district choropleth maps observation frequency to Berlin districts.

11 Source patterns across districts



The heatmap shows which sources dominate specific Berlin districts.

12 Top recorded species in Berlin

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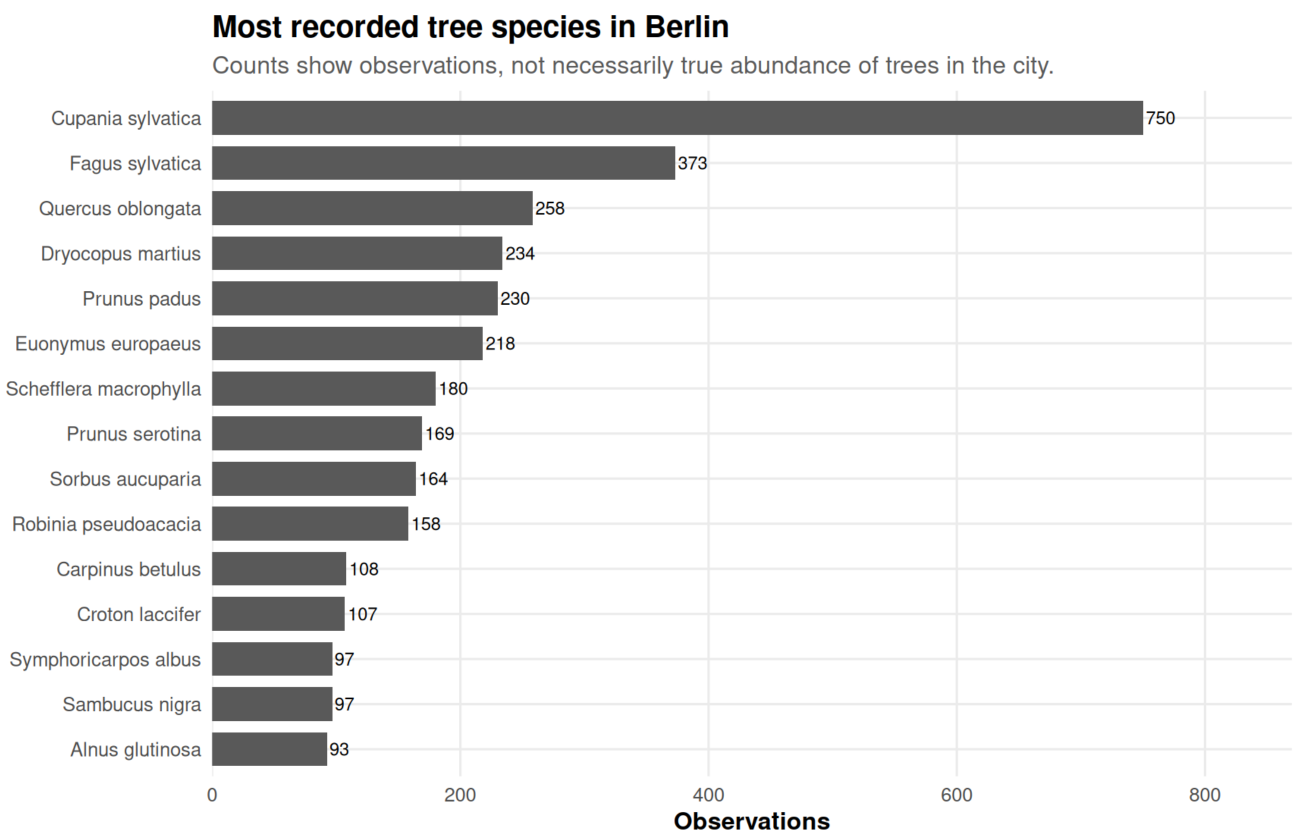
Top recorded species in Berlin

level3_species	observations	share
Cupania sylvatica	750	14.4%
Fagus sylvatica	373	7.2%
Quercus oblongata	258	5.0%
Dryocopus martius	234	4.5%
Prunus padus	230	4.4%
Euonymus europaeus	218	4.2%
Schefflera macrophylla	180	3.5%
Prunus serotina	169	3.3%
Sorbus aucuparia	164	3.2%
Robinia pseudoacacia	158	3.0%
Carpinus betulus	108	2.1%
Croton laccifer	107	2.1%
Sambucus nigra	97	1.9%
Symphoricarpos albus	97	1.9%
Alnus glutinosa	93	1.8%



The most observed species may reflect both Berlin vegetation and observation behavior.

13 Interpretation notes

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The key distinction in this Berlin analysis is between **observation frequency** and **tree abundance**. A district with many records may have many trees, but it may also have more active observers, better platform coverage, or more accessible public spaces. Similarly, a frequently recorded species is not automatically the most common biological species in Berlin; it is the most common species in this dataset.

District-level maps are therefore best interpreted as maps of **recorded tree observations**, not direct ecological census maps.

14 Reproducibility notes

For stable district maps, save a Berlin district GeoJSON file as:

```
berlin_districts.geojson
```

or:

```
data/external/berlin_districts.geojson
```

If no district file is found, this report tries to download one automatically. If the download fails, district-level visualizations are replaced with a fallback note and a Berlin bounding-box density map.